Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/07/2009
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	Comparative Analysis of Fe-Induced Autosomal Mutations in Murine Tissue and Cell Lines		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	a_kronenberg@lbl.gov	**Fax:**	FY 510-486-4475
PI Organization Type:	GOVERNMENT	**Phone:**	510-486-6449
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	1 Cyclotron Road, Building 977-180		
PI Address 2:			
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	05/19/2003	**End Date:**	03/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 03/31/2008 per PI (6/07)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Turker, Mitchell (Oregon Health and Sciences University) Gauny, Stacey (Lawrence Berkeley National Laboratory) Connolly, Lanelle (Oregon Health and Sciences University) Dan, Cristian (Oregon Health and Science University) Kwoh, Ely (Lawrence Berkeley National Laboratory)		
Grant/Contract No.:	T-403-X		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Cancer induction in humans involves a variety of genetic and epigenetic alterations, including widespread loss of expression of genes located primarily on autosomes. Currently, no in vivo data are available to determine the risks of mutagenesis induced by space radiations at an autosomal locus. This study will determine the mutagenic risks of Fe ions using a model autosomal locus (Aprt) in the B6D2 mouse. Four aims are proposed: Aim 1 tests the hypothesis that Fe ions induce autosomal mutations at the Aprt locus in B6D2 kidney epithelial cells exposed in vivo. Aim 2 tests the hypothesis that the exposure to Fe ions elicits primarily multi-locus deletions inclusive of Aprt in B6D2 kidney epithelial cells exposed in vivo. Aim 3 compares Fe-induced mutant frequencies in B6D2 kidney epithelial cells exposed in vivo or in vitro to test the hypothesis that the primary determinant of mutagenic risk is the energy deposited in the cells and not potential modulating effects of the cellular microenvironment. Aim 4 tests the hypothesis that the primary determinant of mutational spectrum is the energy deposited in the cells at risk by comparing the spectrum of Aprt mutants arising after in vitro exposure of B6D2 kidney epithelial cell lines to Fe ions with the spectrum of Aprt mutants obtained after in vivo exposure of kidney epithelial cells (Aim 2). Most literature on the biological effects of heavy ions describes results obtained with in vitro assays, yet considerable uncertainty exists as to how well these assays predict in vivo responses, particularly for endpoints relevant to carcinogenesis. Upon completion of the proposed work, NASA should be in a better position to determine whether cellular assays can be relied upon with a high degree of confidence, or whether greater emphasis should be placed on the establishment and testing of animal models. The goal of this study is to determine whether the tissue microenvironment modifies the mutagenic response of epithelial cells to charged particle radiation of importance in spaceflight. Mutant frequencies and mutant spectra will be evaluated in isogenic kidney epithelial cells exposed either in vitro or in vivo. Little information is available on the mutagenic effects of charged particle radiations in epithelial cell systems. The primary goal of this work is to establish such data and to determine how or if the cellular microenvironment modulates the expression of mutagenic damage.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will be useful for the assessment of the importance of the cellular microenvironment on the frequency and types of mutations induced at autosomal loci following exposure to ionizing radiation, with a particular emphasis on densely ionizing radiations such as alpha particles. These studies may also be of interest to the medical community as it considers the use of charged particles more widely in radiotherapy applications.
Task Progress:	This project focused on the cytotoxic and mutagenc effects of exposure to 1 GeV/amu Fe ions. The goals were to compare and contrast results obtained for isogenic kidney epithelial cells exposed either in vitro or in vivo to determine the influence of the tissue microenvironment on these genotoxic endpoints. In this final review period, we completed the collection of data from in vitro and in vivo analyses of cytotoxicity, mutant frequencies and mutant spectra in kidney epithelium from B6D2 F1 mice heterozygous for the Aprt locus. The results of with different groups of cells established from three primary kidney explants indicated that in vitro exposure followed by immediate plating was associated with exponential cell killing (dose range 0-3 Gy). Surviving fractions averaged between 10-15% at 2 Gy when cells were seeded in colony forming assays immediately after irradiation. Cytotoxicity was also observed in kidney cells explanted from mice two to four months post-irradiation, but to a lesser extent. A dose-dependent decrement in colony formation was measured, with a 52% reduction in cloning efficiency at the highest dose of 2 Gy compared with parallel un-irradiated controls. Incubation in vivo for periods up to nine months post-irradiation was associated with a diminution of cytotoxicity at lower doses (up to 1.5 Gy) as compared with results obtained after the shorter in vivo incubation period. No significant sparing was noted for cell killing after prolonged incubation in vivo after exposure at the highest dose of 2 Gy, where there was a 38% reduction in cloning efficiency at the late time point. The studies of Aprt mutant frequency as a function of dose were completed. Mutant frequencies increased as a linear function of dose for in vitro exposures to 1 GeV/amu Fe ions, with plating within a few days post-irradiation. Similar results were obtained for mutants derived from irradiated kidneys at both time points surveyed ($p<0.001$ for in vivo exposures; $p<0.001$ for in vitro exposures). Background Aprt mutant fractions in vivo ranged between 2 x 10-5 and 5.8 x 10-4. Background mutant fractions were often higher for cells used for in vitro studies due to the need to irradiate large cell numbers to ensure good statistics as a consequence of the high initial toxicity of Fe ions for cells irradiated in vitro that are assayed shortly after the exposure. Nonetheless, the average mutant fraction for a dose of 2 Gy of Fe ions administered in vitro was approximately 3x the average background frequency (or an increase of 1.5x per Gy) while we observed a 1.49x fold increase in mutant fraction per Gy of Fe ions administered in vivo. No differences in mutant frequency were apparent for kidneys harvested 3-4 months post-irradiation incubation in vivo as compared with kidneys harvested after 8-9 months post-irradiation incubation in vivo. Mutant spectra analyses were completed for individual Aprt mutants picked from in vitro exposures to 2 Gy of Fe ions and for a parallel group of spontaneously arising mutants. The results indicate a clear difference in mutant spectra that is highly statistically significant ($p<0.001$) with an enrichment for large interstitial deletion mutants and for mutants arising via mitotic recombination. Large interstitial deletions and mitotic recombination events were also elevated in a statistically significant manner for Aprt mutants collected from Fe ion exposed kidneys, as were mutants with discontinuous loss of heterozygosity. This latter class of events was elevated at both time points after irradiation in situ, but was not evident in cells exposed in culture. In sum, this study demonstrated that Fe ions are potent inducers of cell death, with a relative biological effectiveness in vitro of 2.0 as compared with 160 kVp x-rays. For Fe ion-induced toxicity after in vivo exposures, the results suggest that there is repair and/or remodeling in the intact organ that helps ameliorate the toxic effects, except at the highest dose employed in vivo. Mutant frequencies established in vitro and in vivo were similar, suggesting that the events that lead to mutations are not strongly influenced by the tissue microenvironment. There were also substantial similarities in Fe-ion induced mutant spectra, with most events requiring the resolution of DNA double-strand breaks. There was a suggestion that some unique mutational events are detected in vivo, in particular mutants that reflect multiple rearrangements of the affected chromosome.

Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Kronenberg A, Gauny SS, Connolly L, Turker M. "Genotoxic effects of 1 GeV/amu Fe ions in mouse kidney epithelial cells." Presented at the 51st Annual Meeting of the Radiation Research Society, St. Louis, MO, April 2004. Proceedings, 51st Annual Meeting of the Radiation Research Society, St. Louis, MO, April 2004. , Apr-2004
Abstracts for Journals and Proceedings	Kronenberg A, Gauny SS, Connolly L, Turker M. "Genotoxic effects of 1 GeV/amu Fe ions in mouse kidney epithelial cells." Presented at the Third International Workshop on Space Radiation Research, Port Jefferson, NY, May 2004. Third International Workshop on Space Radiation Research, Port Jefferson, NY, May 2004. , May-2004
Abstracts for Journals and Proceedings	Durante M, Kronenberg A. "Ground-based iron ion biology and physics." Presented at the 35th International Congress on Space Research (COSPAR), Paris, France, July 2004. 35th International Congress on Space Research (COSPAR), Paris, France, July 2004. , Jul-2004
Abstracts for Journals and Proceedings	Kronenberg A, Gauny SS, Connolly L, Turker M. "Impact of the tissue microenvironment on cell killing and autosomal mutagenesis in murine cells exposed to 1 GeV/amu Fe ions." 16th NASA Space Radiation Health Investigators' Workshop Proceedings, Port Jefferson, NY, May 2005. 16th NASA Space Radiation Health Investigators' Workshop Proceedings, 2005. , May-2005
Abstracts for Journals and Proceedings	Kronenberg A. "Exploring the roles of physics and biology in mediating charged particle radiation responses in mammalian cells and tissues." European Radiation Research Meeting, Leicester, England, September 2005. Proceedings of the European Radiation Research Society Meeting, Leicester, England, September 2005. , Sep-2005
Abstracts for Journals and Proceedings	Kronenberg A, Gauny SS, Connolly L, Turker M. "Impact of the tissue microenvironment of cell killing and autosomal mutagenesis in murine epithelial cells exposed to 1 GeV/amu Fe ions." 52nd Annual Meeting of the Radiation Research Society, Denver, CO, October 2005. Proceedings of the 52nd Annual Meeting of the Radiation Research Society, Denver, CO, October 2005. , Oct-2005
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Connolly L, Dan C, Turker M. "Comparative analysis of Fe ion-induced mutations in mouse tissue and cells." Presented at the 4th International Workshop on Space Radiation Research and 17th NASA Space Radiation Health Investigators' Meeting, Moscow and St. Petersburg, June 2006. 4th International Workshop on Space Radiation Research and 17th NASA Space Radiation Health Investigators' Meeting, Moscow and St. Petersburg, June 2006. , Jun-2006
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Dan C, Connolly L, Turker M. "Comparative analysis of Fe ion-induced mutations in murine tissue and cells." Presented at the 36th COSPAR Scientific Assembly, Beijing, China, July 2006. 36th COSPAR Scientific Assembly, Beijing, China, July 2006. , Jul-2006
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Connolly L, Dan C, Turker M. "Impact of hte tissue microenvironment on the induction of autosomal mutations in murine kidney epithelial cells irradiated with 1 GeV/amu Fe ions." Presented at the COSPAR Colloquium on the Mutagenic Consequences of the Space Environment, Xi'an China, July 2006. COSPAR Colloquium on the Mutagenic Consequences of the Space Environment, Xi'an China, July 2006. , Jul-2006
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Connolly L, Dan C, Turker M. "Toxic and mutagenic effects of Fe ions for kidney epithelial cells exposed in vivo and in vitro." Presented at the 53rd Annual Meeting of the Radiation Research Society, Philadelphia, PA, November 2006. 53rd Annual Meeting of the Radiation Research Society, Philadelphia, PA, November 2006. , Nov-2006
Abstracts for Journals and Proceedings	Sudo H, Stampfer M, Barcellos-Hoff MH, Kronenberg A. "Induction of genomic instability in finite lifespan human mammary epithelial cells by exposure to sparsely or densely ionizing radiation." Presented at the 53rd Annual Meeting of the Radiation Research Society, Philadelphia PA, November 2006. 53rd Annual Meeting of the Radiation Research Society, Philadelphia, PA, November 2006. , Nov-2006
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Wiese C, Dan C, Connolly L, Turker M. "Common findings regarding charged particle exposure to murine and human cells and tissues." Presented at the NASA Workshop on Models of Radiation Risk, Dallas, TX, March 2007. NASA Workshop on Models of Radiation Risk, March 2007. , Mar-2007
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Wiese C, Dan C, Turker M. "Charged particle-induced mutagenesis and genomic instability in vitro and in vivo." Presented at the 13th International Congress of Radiaiton Research, San Francisco, CA, July 2007. 13th International Congress of Radiation Research, San Francisco, CA, July 2007. , Jul-2007
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Wiese C, Dan C, Turker M. "Comparative analysis of Fe ion-induced autosomal mutations in mammalian models in vitro and in vivo." Presented at the 18th Annual Space Radiation Investigators' Workshop, Rohnert Park, CA, July 2007. 18th Annual Space Radiation Investigators' Workshop, Rohnert Park, CA, July 2007. , Jul-2007
Abstracts for Journals and Proceedings	Sudo J, Garbe JC, Stampfer MR, Barcellos-Hoff MH, Kronenberg A. "Genomic instability in finite lifespan human mammary epithelial cells exposed to x-rays or Fe ions." Presented at the 18th Annual Space Radiation Investigators' Workshop, Rohnert Park, CA, July 2007. 18th Annual Space Radiation Investigators' Workshop, Rohnert Park, CA, July 2007. , Jul-2007
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Dan C, Turker M. "Comparative analysis of charged particle-induced mutations in murine cells and tissues." Presented at the NASA Human Research Program Investigators' Workshop, Houston, TX, February 2008. NASA Human Research Program Investigators' Workshop, Houston, TX, February 2008. , Feb-2008
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Dan C, Turker M. "Autosomal mtuagenesis after charged particle exposure of mouse kidney epithelial cells." Presented at the NASA Space Radiation Health Investigators' Workshop, Philadelphia, PA, June 2008. NASA Space Radiation Health Investigators' Workshop, Philadelphia, PA, June 2008. , Jun-2008

Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Dan C, Turker M. "Comparative analysis of charged particle-induced autosomal mutations in murine cells and tissues." Presented at the 27th COSPAR Scientific Assembly, Montreal, Canada, July 2008. 27th COSPAR Scientific Assembly, Montreal, Canada, July 2008. , Jul-2008
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Dan C, Turker M. "Charged particle-induced autosomal mutations and the influence of the tissue microenvironment." Presented at the 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 2008. 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 2008. , Sep-2008
Articles in Peer-reviewed Journals	Durante M, Kronenberg A. "Ground-based research with heavy ions for space radiation protection." Adv Space Res. 2005;35(2):180-4. PMID: 15934192 , Jul-2005
Articles in Peer-reviewed Journals	Kronenberg A. et al. Organizing Committee, American Statistical Association Conference on Radiation and Health Pacific Grove, California, June 18–21, 2006. "Radiation Research: State of the science 20 years after Chernobyl." Radiation Research. 2007 Mar;167(3):338-60, 2007. http://dx.doi.org/10.1667/RR0890.1 , Mar-2007
Articles in Peer-reviewed Journals	Chyall LH, Gauny SS, Kronenberg A. "Characterization of a TK6-Bcl-X(L) gly-159-ala human lymphoblast clone." Journal of Undergraduate Research (U.S. Department of Energy) 2006;6:35-9, 2006. http://www.scied.science.doe.gov/scied/JUR_v6/PDFs/Chyall.pdf , Oct-2006
Articles in Peer-reviewed Journals	Sudo H, Garbe J, Stampfer MR, Barcellos-Hoff MH, Kronenberg A. "Karyotypic instability and centrosome aberrations in the progeny of finite life-span human mammary epithelial cells exposed to sparsely or densely ionizing radiation." Radiat Res. 2008 Jul;170(1):23-32. PMID: 18582160 , Jul-2008